ARPIT SHARMA

Full Stack AI Engineer | Backend & LLM Systems
Delhi, India
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PROFESSIONAL SUMMARY

B.Tech 2026 graduate specialising in AI-native full-stack engineering — with hands-on experience building RAG pipelines, agentic workflows, and LLM-integrated systems. Strong backend foundation in Java, Spring Boot, Python, Django, and REST APIs. Completed an AI internship working on SFT and RLHF pipelines; deeply curious about prompt engineering, AI evaluation, and production-grade intelligent systems. Ready to ship real AI-native features on enterprise codebases from day one.

SKILLS SUMMARY

Languages & Frameworks: Java, Python, SQL; Spring Boot, Django, FastAPI, REST APIs, MVC Architecture
AI & LLM: LLM integration, RAG pipelines, Prompt engineering, LangChain, LangGraph, CrewAI, Embeddings, Structured outputs, LLM Tool-Calling, RLHF / SFT, AI output evaluation & guardrails
Frontend & UI: Streamlit, React, JavaScript, HTML/CSS, REST API integration
Databases: MySQL, MongoDB, FAISS, Vector DB concepts
Cloud & DevOps: AWS (EC2, S3), Docker, Git, CI/CD Pipelines
Testing & Quality: Postman, Unit & Integration Testing, API Testing, AI output validation & guardrails
AI Tools: GitHub Copilot, Claude, OpenAI API, Google Gemini API, Jupyter Notebook
CS Fundamentals: DSA, OOP, OS, DBMS, Computer Networks, SDLC

EDUCATION

Noida Institute of Engineering and Technology Jul 2022 – Jul 2026
B.Tech in Information Technology | CGPA: 8.28

INTERNSHIP / EXPERIENCE

Ethara AI (Green Rider Technology LLP) Jan 2026 – Mar 2026 *Artificial Intelligence Intern* — Remote

- Contributed to **LLM post-training workflows**, supporting data curation and annotation for **SFT (Supervised Fine-Tuning)** and **RLHF (Reinforcement Learning from Human Feedback)** pipelines.
- Reviewed and refined AI-generated outputs, improving **response accuracy, consistency, and alignment** with safety and quality standards — directly relevant to AI evaluation and guardrails.
- Developed hands-on familiarity with **prompt evaluation**, structured output review, and iterative refinement of model responses for production readiness.

NIET IIC & AICTE IDEA Lab May 2023 – Jul 2023 *Innovation Intern* — Noida, Uttar Pradesh

- Designed and built an **IoT-enabled Smart Lamp** integrating embedded systems with real-time backend data handling.
- Developed **Java-based backend components** for processing and visualizing live sensor data, applying REST and data-flow principles.

PROJECTS

AI Workflow Orchestration Engine — General-Purpose Multi-Agent Platform | github.com/asomlt/AI-Workflow-Orchestration-Engine
Python, CrewAI, LangGraph, FastAPI, OpenAI GPT / Claude, Docker, LLM Tool-Calling

- Designed a **domain-agnostic multi-agent orchestration platform** using CrewAI with pluggable agent roles (planner, generator, validator, executor) — enabling rapid deployment across content, data, and enterprise workflow use cases.
- Implemented **LangGraph-based stateful workflow graphs** with conditional branching, iterative validation loops, and retry logic — ensuring robust, production-grade agentic pipeline execution.
- Designed **LLM tool-calling patterns and structured output schemas** enabling agents to invoke external APIs, parse results, and pass typed data between pipeline stages with full traceability.

- Exposed the engine via **FastAPI REST endpoints with Docker containerisation**; tracked per-run token cost, latency, and agent execution steps to support cost-accuracy optimisation in production deployments.

Legal Intelligence Agent — Agentic RAG System with Smart Query Routing | github.com/asomlt/Legal-Intelligence-Agent

Python, LangChain, LangGraph, Google Gemini API, FAISS, Vector Embeddings, FastAPI

- Designed an **agentic query router using LangGraph** that dynamically decides between RAG retrieval, direct LLM response, or human escalation — optimising accuracy and cost per query type.
- Built a **semantic retrieval system using FAISS vector embeddings** over constitutional laws and case data; benchmarked **RAG vs vector-only retrieval** to minimise hallucinations and maximise retrieval relevance.
- Integrated **Google Gemini API with structured output enforcement** to generate citation-backed legal responses, classify case types (intentional vs. unintentional, domestic vs. international), and apply guardrails against unsafe outputs.
- Implemented a **self-refinement feedback loop** where the agent evaluates its own response confidence, re-queries with refined prompts on low-confidence answers, and logs all interactions for continuous quality improvement.

RepoDoc AI — AI Dev Assistant with RAG, Eval & Monitoring | github.com/asomlt/RepoDoc-AI

Python, LangChain, FastAPI, FAISS, OpenAI Embeddings, Streamlit, LLM Eval Metrics

- Architected an **end-to-end RAG pipeline** ingesting GitHub repository content with recursive chunking strategy, FAISS vector indexing, and context-aware retrieval for LLM-based code documentation generation.
- Engineered **structured prompt templates and schema-enforced outputs** via FastAPI to generate consistent, multi-language documentation (Python, JavaScript, Java) from raw source code using LLM inference.
- Built an **AI evaluation layer** scoring each LLM response on **relevance, faithfulness, and completeness** — flagging hallucinations and triggering re-generation when quality thresholds are not met.
- Implemented a **monitoring dashboard (Streamlit)** tracking query latency, average quality scores, and token cost per request — enabling cost-accuracy optimisation aligned with production AI Ops standards.

LinkedIn-Inspired Social Platform | github.com/asomlt/opportunities-LinkedIn_backend

Java, Spring Boot, REST APIs, WebSocket (STOMP), MySQL, MongoDB, JWT, Spring Security

- Engineered a **modular REST API backend using Spring Boot** covering user management, connections, posts, and feed — following API-first, layered architecture patterns applicable to enterprise client codebases.
- Implemented **real-time messaging using WebSocket (STOMP protocol)** enabling bidirectional chat between users — demonstrating event-driven, low-latency communication architecture.
- Secured all endpoints with **JWT authentication and Spring Security RBAC** — enforcing role-based access for users, admins, and moderators across all platform features.
- Designed a **dual-database architecture (MySQL + MongoDB)** — MySQL for structured relational data (users, connections) and MongoDB for flexible content (posts, messages) — demonstrating polyglot persistence design.

CERTIFICATIONS & LEARNING

- **Google Cloud – Generative AI Learning Path** — Generative AI, embeddings, vector search, and RAG pipelines using Vertex AI (Gemini); hands-on labs aligned with LLM integration workflows.
- **DeepLearning.AI – ChatGPT Prompt Engineering for Developers** — Prompt design, chaining, system messages, and structured outputs for real-world LLM applications.
- **Microsoft – Azure AI Fundamentals (AI-900)** — AI workloads, NLP, and computer vision on Azure; enterprise-grade AI deployment credentialing.
- **AWS – Cloud Practitioner Essentials** — Cloud architecture, deployment, CI/CD, and container-based scaling; complements existing AWS (EC2, S3) hands-on experience.
- **Hugging Face – Transformers Course** — Transformers, embeddings, and fine-tuning fundamentals; deepens understanding of LLM internals applicable to SFT/RLHF workflows.

ADDITIONAL HIGHLIGHTS

- **Genuine AI Curiosity:** Hands-on experience with LLM workflows (SFT, RLHF) and exploring LangChain, RAG, and agentic systems independently.
- **AI Tooling:** Daily use of GitHub Copilot and Claude for AI-augmented development; actively building projects with AI coding assistants.
- **Communication & Collaboration:** Presented technical projects to faculty panels; experience in cross-functional innovation lab team.
- **Graduating 2026** (B.Tech IT) — immediately available for full-time associate-level engineering roles.